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groupe 3

Final sheet

1- Validation testing → To demonstrate to the developer & the system customer that the software meets its requirements, a successful test shows that the system operates as intended.
Defect Testing → To discover faults or defects in the software where its behavior is incorrect or not in conformance with its specification

2- Test driven development is an approach to Program development in which you inter-leave testing and code development
its benefits are Code Coverage where all code written has a least one test, Regression testing where it's developed incrementally as a Program is developed, Simplified debugging when a test fails the newly written code needs to be checked and modified, System documentation where tests are a form of documentation that describe what the code should be doing

3- Equivalence Partition is classes where input data and output results often fall into different classes where all members of a class are related
Test Cases are chosen from each Partition

4- Development testing where the system is tested during development to discover bugs and defects
Release testing where a separate testing team test a complete version of the system before it's released to users.
User testing, where users or Potential users of a system test the system in their own environment

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5- Re-engineering takes place after a system has been maintained for some time & maintenance costs are increasing. you use automated tools to Process and re-engineer a legacy system to create a new system that is more maintainable

- Refactoring is a continuous Process of improvement throughout the development and evolution Process. it is intended to avoid the structure & code degradation

6- Continuing change → A Program that is used in real-world environment must necessarily change, or else become progressively less useful in that environment

* Continuing growth → The functionality offered by systems has to continually increase to maintain user satisfication

* Increasing Complexity → As an evolving Program changes its structure tends to become more complex. Extra resources must be devoted to Preserving & simplifying the structure

7- Interface misuse → A Calling component calls another component & makes an error in its use of its interface

Interface misunderstanding → A Calling Component embeds assumptions about the behavior of the called component which are incorrect

Timing errors → The Called & the Calling Component operate at different speeds & out-of-date information is accessed

8- Alpha testing → Users of the software work with the development team to test the software at the developer's site

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Beta testing → A release of the software is made available to users to allow them to experiment & to raise Problems that they discover with the system developers

Acceptance testing → Customers test a system to decide whether or not it's ready to be accepted from the system developers & deployed in the customer environment. Primarily for custom systems.

9. Duplicate code → same or very similar code may be included at different places in a Program. This can be removed & implemented as a single method

- Long methods → if a method is too long, it should be redesigned as a number of shorter methods

- Switch statements → These often involve duplication where the switch depends on the type of a value. The switch statements may be scattered around a Program.

10- Organisations that rely on legacy systems must choose a strategy for evolving these systems, the four strategies are;

Scrap the system completely and modify business Processes so that it's no longer required

Continue maintaining the system

Transform the system by re-engineering to improve it's maintainability

Replace the system with a new system.

Strategy chosen should depend on the system quality & its business value

HCQ

1- C
2- B
3- C
4- C
5- C
6- D
7- A
8- D
9- C
10- C

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